1. (1) $A\bar{B}\bar{C}$

(2) $A\bar{B}\bar{C} \cup \bar{A}B\bar{C} \cup \bar{A}\bar{B}C$

(3) $A \cup B \cup C$

(4) $\bar{A}\bar{B}\bar{C} \cup A\bar{B}\bar{C} \cup \bar{A}B\bar{C} \cup A\bar{B}\bar{C}$

(5) $\bar{A}\bar{B}\bar{C}$

(6) $\bar{A}(B \cup C)$

3. (1) $A \cup B = \{x \mid 1 \leq x \leq 4\}$

(2) $AB = \{x \mid 2 < x \leq 3\}$

(3) $\bar{A} = \{x \mid 0 \leq x < 1 \text{ 或 } 3 < x \leq 5\}$

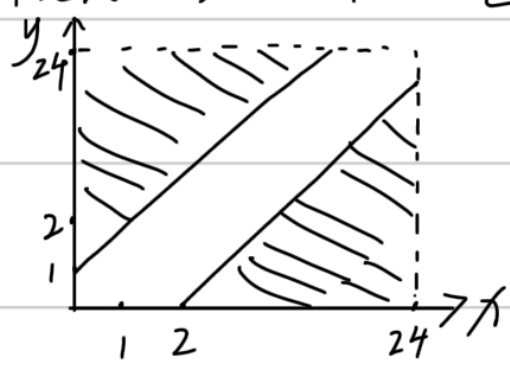
(4) $A - B = \{x \mid 1 \leq x \leq 2\}$

5. 两面涂有红色的正方体的个数： $8 \times 12 = 96$

$$P = \frac{96}{1000} = \frac{12}{125}$$

7. 设甲船停靠1小时，x 时停靠，乙船停靠2小时，y 时停靠

甲先到：$y \geq x+1$　　乙先到：$x \geq y+2$



$$P = 1 - \frac{\frac{1}{2} \times 22^2 + \frac{1}{2} \times 23^2}{24 \times 24} = \frac{139}{1152}$$

9. (1) ∵ $P(AC)=0$

∴ A, C 不可能同时发生

∴ $P(ABC)=0$

(2) $P(A\cup B\cup C)=P(A)+P(B)+P(C)-P(AB)-P(AC)-P(BC)+P(ABC)$

$=3\times\frac{1}{4}-2\times\frac{1}{8}=\frac{1}{2}$

(3) $P(\bar{A}\bar{B}\bar{C})=1-P(A\cup B\cup C)=\frac{1}{2}$

11.

不放回: $\Omega_1=C_{10}^2=45$

放回: $\Omega_2=10\times10=100$

(1) $P(A_1)=\frac{C_4^2}{45}=\frac{2}{15}$ $\quad P(A_2)=\frac{4\times4}{100}=\frac{4}{25}$

(2) $P(B_1)=\frac{C_4^1\times C_6^1}{2\times45}=\frac{4}{15}$ $\quad P(B_2)=\frac{4\times6}{100}=\frac{6}{25}$

(3) $P(C_1)=\frac{C_4^1\times C_6^1}{45}=\frac{8}{15}$ $\quad P(C_2)=\frac{2\times4\times6}{100}=\frac{12}{25}$

(4) $P(D_1)=\frac{6\times4+4\times3}{2\times45}=\frac{2}{5}$ $\quad P(D_2)=\frac{10\times4}{100}=\frac{2}{5}$

13. 设 $A=\{4$个人都在不同月份出生$\}$

$\Omega=12^4$

$P(A)=\frac{C_{12}^4\times4!}{12^4}=\frac{11\times10\times9}{12^3}=\frac{55}{96}$

$P=1-P(A)=\frac{41}{96}$

∴ 至少2个人同月出生概率为$\frac{41}{96}$

15. $\Omega = C_{10}^4 \times 4!$

尾数为偶的情况共：$C_5^1 \times C_9^3 \times 3!$ 个

首位为0，尾数为偶数的情况共：$C_4^1 \times C_8^2 \times 2!$

$\therefore P = \dfrac{5 \times C_9^3 \times 3! - C_4^1 \times C_8^2 \times 2!}{C_{10}^4 \times 4!} = \dfrac{41}{90}$

17. $P(\bar{B} \mid A \cup B) = \dfrac{P(\bar{B}(A \cup B))}{P(A \cup B)} = \dfrac{P(\bar{B}A \cup \bar{B}B)}{P(A \cup B)} = \dfrac{P(\bar{B}A) + P(\bar{B}B)}{P(A \cup B)}$

$P(\bar{B}B) = 0$

$P(A \cup B) = P(A) + P(B) - P(AB) = 0.7 + 0.4 - 0.2 = 0.9$

$P(\bar{B}A) = P(A) - P(AB) = 0.7 - 0.2 = 0.5$

$\therefore P(\bar{B} \mid A \cup B) = \dfrac{5}{9}$

19. $\Omega = 100 \times 99 \times 98$

$P = \dfrac{7 \times 6 \times 93}{100 \times 99 \times 98} = \dfrac{31}{7700}$

21. 设 $A_i = \{$抽到 i 级射手$\}$，$P(B)$ 为一次射中10环

$P(A_1) = \dfrac{4}{20} = \dfrac{1}{5}$　$P(A_2) = \dfrac{2}{5}$　$P(A_3) = \dfrac{7}{20}$　$P(A_4) = \dfrac{1}{20}$

$P(B) = \sum_{i=1}^{4} P(B \mid A_i) P(A_i)$

$= 0.9 \times \dfrac{1}{5} + 0.7 \times \dfrac{2}{5} + 0.5 \times \dfrac{7}{20} + 0.2 \times \dfrac{1}{20}$

$= 0.645$

23. 设 $A=\{$抽到A用户$\}$，$B=\{$抽到B用户$\}$，$C=\{$抽到C用户$\}$

$D=\{$好评$\}$

(1) $P(A)=\frac{4}{9}$ $P(B)=\frac{1}{3}$ $P(C)=\frac{2}{9}$

$P(D)=P(D|A)\cdot P(A)+P(D|B)\cdot P(B)+P(D|C)\cdot P(C)$

$=\frac{4}{9}\times 0.8+\frac{1}{3}\times 0.6+\frac{2}{9}\times 0.7$

$=\frac{32}{45}$

(2) $P(B|D)=\frac{P(BD)}{P(D)}=\frac{P(D|B)\cdot P(B)}{P(D)}=\frac{\frac{1}{3}\times 0.6}{\frac{32}{45}}=\frac{9}{32}$

25. $\because P(A|B)=P(A|\bar{B})$

$\therefore P(A|B)P(B)=P(A|\bar{B})P(B)$

$\therefore P(AB)=P(A|\bar{B})(1-P(\bar{B}))$

$\therefore P(AB)=P(A|\bar{B})-P(A\bar{B})$

$\because P(A\bar{B})=P(A)-P(AB)$

$\therefore P(AB)=P(A|\bar{B})-P(A)+P(AB)$

$\therefore P(A)=P(A|\bar{B})$

$\therefore P(A)P(B)=P(AB)$

$\therefore$ A，B相互独立

27. 设 $A=\{$甲射中$\}$，$B=\{$乙射中$\}$，$C=\{$丙射中$\}$，$D=\{$目标被击中$\}$

$P(A)=\frac{4}{5}$，$P(B)=\frac{2}{3}$，$P(C)=\frac{3}{4}$，A,B,C 相互独立

$P(D)=1-P(\bar{A}\bar{B}\bar{C})=1-P(\bar{A})\cdot P(\bar{B})\cdot P(\bar{C})=1-\frac{1}{5}\times\frac{1}{3}\times\frac{1}{4}=\frac{59}{60}$

29. 系统Ⅰ：设 $A=\{$Ⅰ能正常工作$\}$

设每条支路正常的概率为 P_0

$P_0=p^n$

$P(A)=2P_0(1-P_0)+P_0^2=2P_0-P_0^2=p^n(2-p^n)$

系统Ⅱ：设 $B=\{$Ⅱ能正常工作$\}$

$C=\{$每个节点至少有一个正常$\}$

$P(C)=1-(1-p)^2$

$P(B)=P^n(C)=[1-(1-p)^2]^n=(2p-p^2)^n=p^n(2-p)^n$

$\therefore \frac{P(A)}{P(B)}=\frac{p^n(2-p^n)}{p^n(2-p)^n}=\frac{2-p^n}{(2-p)^n}$

$n=1$ 时，$P(A)=P(B)$，$n=2$ 时，$P(A)<P(B)$

设 $n=k$ 时，$2-p^k<(2-p)^k$

$\therefore (2-p)^{k+1}=(2-p)(2-p)^k>(2-p)(2-p^k)=4-2p^k-2p+p^{k+1}$

又 $\because 4-2p^k-2p+p^{k+1}-2+p^{k+1}=2-2p^k-2p+2p^{k+1}=2(1-p)(1-p^k)>0$

$\therefore (2-p)^{k+1}>2-p^k$

$\therefore$ 由数学归纳法，$n\geq 2$ 时 $P(A)<P(B)$ 恒成立

$\therefore$ 系统Ⅱ可靠性大

1. (1) $\Omega = 10!$

$$P_1 = \frac{5! \times 2^5}{10!} = \frac{1}{945}$$

(2) 设 $A = \{$妻子在丈夫两面$\}$

$$P(A) = \frac{1}{2}$$

$$\therefore P_2 = P(A) = \frac{1}{2^5} = \frac{1}{32}$$

(3) $\Omega = 9!$

$$P_3 = \frac{4! \times 2^5}{9!} = \frac{2}{945}$$

3. (1) 设 $A_i = \{$损坏了 i 个元件$\}$，$B = \{$仪器故障$\}$

$$P(A_1) = 3 \times 0.2 \times 0.8^2 = \frac{48}{125}$$

$$P(A_2) = 3 \times 0.2^2 \times 0.8 = \frac{12}{125}$$

$$P(A_3) = 0.2^3 = \frac{1}{125}$$

$$P(B) = \sum_{i=1}^{3} P(B|A_i) P(A_i)$$

$$= 0.25 \times \frac{48}{125} + 0.6 \times \frac{12}{125} + 0.95 \times \frac{1}{125}$$

$$= \frac{403}{2500}$$

(2) $$P(A_2|B) = \frac{P(A_2 B)}{P(B)} = \frac{P(B|A_2)P(A_2)}{P(B)} = \frac{0.6 \times \frac{12}{125}}{\frac{403}{2500}} = \frac{144}{403}$$

5. (1) I：设 $A=\{支路不正常\}$

$P(A)=1-p^2$

$P_1=\left[1-(1-p^2)^2\right]p=2p^3-p^5$

II：$P_2=2p(1-p)\left[p+(1-p)p^2\right]+p^2\left[1-(1-p)^2\right]$

$=(2p-2p^2)(p+p^2-p^3)+(p^2-p^4+2p^3-p^2)$

$=2p^5-4p^4+2p^2-p^4+2p^3$

$=2p^5-5p^4+2p^3+2p^2=p^2(2p^3-5p^2+2p+2)$

∴ I正常工作概率为 $2p^3-p^5$

II正常工作概率为 $p^2(2p^3-5p^2+2p+2)$

(2) 设 $B=\{1,2不能正常工作\}$，$C=\{I子统能正常工作\}$

$P(B)=(1-p)^2$，$P(C|B)=p^3$

$$P(B|C)=\frac{P(BC)}{P(C)}=\frac{P(C|B)\cdot P(B)}{P(C)}=\frac{p^3(1-p)^2}{2p^3-p^5}=\frac{(1-p)^2}{2-p^2}$$

∴ 子统I正常工作时，1，2都不能正常工作的概率为 $\frac{(1-p)^2}{2-p^2}$